

BIP-110

Overview

Metadata

Field	Value
BIP	110
Title	Reduced Data Temporary Softfork
Layer	Consensus (soft fork)
Type / status	Specification / Draft
Author	Dathon Ohm
Assigned	2025-12-03
Snapshot date	2026-06-19

What BIP-110 is

- A proposed temporary soft fork
- Target: large arbitrary-data patterns in Bitcoin transactions
- Duration: 52,416 active blocks, roughly one year
- Existing UTXOs created before activation are exempt
- After expiry, the extra restrictions stop applying

Why it exists

- Supporters see inscriptions and similar protocols as abusive data storage
- Their concern is long-term cost for node operators
- The proposal tries to push Bitcoin back toward monetary use
- It is also meant to send a social signal
- The BIP does not claim to make all arbitrary data impossible

Context

Consensus vs relay policy

- Relay policy decides what a node accepts into its mempool and forwards
- Consensus decides whether a block is valid
- BIP-110 is a consensus proposal
- During the active window, enforcing nodes reject blocks with violating transactions
- Non-upgraded nodes validate those same blocks under the old rules

Outputs, pushes, witness

Term	Meaning here
scriptPubKey	The output locking script. If the output is unspent, it lives in the UTXO set.
OP_RETURN	A provably unspendable output type. BIP-110 gives it a separate 83-byte allowance.
OP_PUSHDATA*	Script opcodes that push byte strings onto the stack. BIP-110 caps their payloads at 256 bytes.
Witness data	SegWit data area for signatures, scripts, and script arguments. It is discounted, which made it attractive for inscriptions.

Taproot terms

Term	Meaning here
Key path	Spend a Taproot output with a signature. No script is revealed.
Script path	Spend a Taproot output by revealing one script alternative.
Tapleaf	One script-path alternative inside Taproot.
Taptree	The Merkle tree of Tapleaves committed by the Taproot output.
Control block	Proof that the revealed Tapleaf belongs to the Taptree. Size is $33 + 32 * \text{depth}$.
Annex	Optional Taproot witness element reserved for future use. Currently undefined.

Tapscript hooks

- Tapscript is the script language used for Taproot script-path spends
- `OP_SUCCESS\*` opcodes are reserved upgrade hooks
- Today, encountering `OP_SUCCESS\*` makes validation succeed
- `OP_IF` / `OP_NOTIF` are conditional branches
- BIP-110 restricts these because they are useful places to hide skipped payloads

Proposed changes

Area	Temporary restriction
scriptPubKey	New output scripts over 34 bytes invalid, except OP_RETURN up to 83 bytes
OP_PUSHDATA*	Payloads over 256 bytes invalid, except BIP-16 redeemScript push
Witness arguments	Script argument witness items over 256 bytes invalid
Undefined witness / Tapleaf versions	Spending them invalid
Taproot annex	Witness stacks with an annex invalid
Taproot control block	Control blocks over 257 bytes invalid
Tapscript	OP_SUCCESS* invalid; executed OP_IF / OP_NOTIF invalid

Effects

Main targets

- Large data in spendable output scripts
- Large contiguous script pushes
- Large script argument witness items
- Undefined witness and Tapleaf version spends
- Taproot annex usage
- Very deep Taproot script trees
- Tapscripts using `OP_SUCCESS*`, `executed OP_IF`, or `executed OP_NOTIF`

Practical reading

- The proposal blocks common, visible data-carrier patterns
- It does not block every possible encoding
- It makes some forms of data embedding more fragmented or expensive
- It also restricts some future-upgrade surfaces for one year
- That tradeoff is the center of the technical objection

Wallet and contract impact

- Common payment outputs fit inside the limits
- Taproot key-path spends are not the target
- Some Miniscript-generated Tapleaves may need adjustment
- Large Taptrees may need shallower constructions
- Pre-signed Taproot transactions need special care if they must confirm and spend during the active window

Grandfathering

- Inputs spending UTXOs created before activation are exempt
- New outputs created after activation are checked
- Hidden Taproot script paths matter because they are revealed only when spent
- After expiry, UTXOs of all heights are unrestricted again
- The rule is height-sensitive, not only transaction-format-sensitive

Activation

Deployment parameters

Parameter	Value
Deployment name	reduced_data
Version bit	4
Start time	2025-12-01 00:00:00 UTC
Timeout	NO_TIMEOUT
Max activation height	965,664
Active duration	52,416 blocks
Threshold	1,109 / 2,016 blocks (55%)

State machine

1. **DEFINED**: before start time
2. **STARTED**: miners may signal bit 4
3. **LOCKED_IN**: threshold reached, or forced no later than height 963,648
4. **ACTIVE**: one retarget period after lock-in; new rules enforced
5. **EXPIRED**: activation height plus 52,416 blocks; new rules stop applying

Important dates and heights

When	Height	Meaning
2025-12-01 00:00 UTC	n/a	Start time
2025-12-03	n/a	BIP assigned
Aug 2026	961,632-963,647	Mandatory signaling window
Aug 2026	963,648	Latest lock-in boundary
Sep 2026	965,664	Latest activation height
1 year after activation	activation + 52,416	Expiry

Two activation paths

- Early path: any retarget period reaches 1,109 signaling blocks
- Mandatory path: blocks 961,632 through 963,647 must signal bit 4
- Lock-in to activation is one retarget period, about two weeks
- If activation is at 965,664, expiry is at 1,018,080
- Calendar dates are estimates; heights are the exact boundaries

What is unusual

Standard BIP-9	BIP-110
95% threshold	55% threshold
Can time out into FAILED	NO_TIMEOUT; no normal failed state
No mandatory signaling	Mandatory signaling before maximum activation
ACTIVE is terminal	ACTIVE later becomes EXPIRED
Permanent once active	Temporary: 52,416 active blocks

Compared with prior activations

	SegWit	Taproot	BIP-110
Threshold	95%	90%	55%
Mechanism	BIP9 + BIP91 pressure	Speedy Trial	Modified BIP9 / BIP8-style forcing
Miner veto	Initially yes	Yes	No, after mandatory window
Duration	Permanent	Permanent	Temporary

Sentiment

Supporter frame

- Bitcoin is money, not data storage
- Data embedding imposes costs on node operators
- A temporary restriction buys time
- Consensus can express a social boundary
- Better to resist now than let the pattern become normal

Opponent frame

- Bitcoin should validate rules, not transaction purpose
- Fee-paying valid transactions should remain neutral
- Data can move into other encodings anyway
- Temporary restrictions still create precedent
- A contentious UASF risks splitting social and economic consensus

Public supporters

Person / project	Public signal
Dathon Ohm	BIP-110 author
Luke Dashjr	Original draft/advice credit; Bitcoin Knots maintainer
Bitcoin Knots	Ships RDTS / BIP-110 support
Léo Haf	Contributor to the Knots implementation
Bitcoin Mechanic	Publicly aligned with the pro-BIP-110 / OCEAN / Knots camp
hodlonaut	Published pro-BIP-110 / anti-Core-capture essays linked from bip110.org
Matthew Kratter / Bitcoin University	Repeated public advocacy and run-BIP-110 guides

Public opponents

Person / group	Public signal
Jameson Lopp	Published a long critique calling BIP-110 reckless and likely to fail
Peter Todd	Argued the activation path risks two coins and data remains publishable
Adam Back	Warned about downgrade, user-space breakage, and minority-fork risk
Wang Chun / F2Pool	Publicly rejected BIP-110 signaling
Antoine Riard	Rejected the legal / coercion rationale on the bitcoin-dev thread
Erik Aronesty	Rejected the legal-liability framing and emphasized uncensorability
Taproot Wizards / Udi Wertheimer	Naturally opposed as inscription-oriented builders

Companies and projects

Entity	Read	Caveat
Bitcoin Knots	Supportive	Ships RDTS / BIP-110 support
bip110.org	Supportive	Campaign site, not a neutral standards body
Barefoot Mining	Supportive	Produced a signaling block via Ocean infrastructure
F2Pool	Opposed	Cofounder Wang Chun publicly rejected signaling
Bitcoin Core contributors	Directionally opposed	Relay-policy statement conflicts with BIP-110 philosophy; not a formal BIP vote

No verified public stance

- Major exchanges: Coinbase, Binance, Kraken
- Hardware-wallet vendors: Ledger, Trezor
- Most wallet companies
- Most public Bitcoin treasury companies
- Most payment apps and custodians
- Most mining pools